

A Survey on Geometric and Temporal Deep Learning Architectures for Financial Time Series Forecasting

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Abstract— The prediction of financial time series has historically been dominated by linear econometric models and, more recently, Recurrent Neural Networks (RNNs) such as Long Short-Term Memory (LSTM) units. While effective for sequence modeling, these “Euclidean” architectures fundamentally treat financial assets as isolated entities, neglecting the complex, dynamic web of spatial correlations that drive systemic market behavior. This survey explores the emerging paradigm of Geometric Deep Learning in finance, which posits that market data resides on irregular, non-Euclidean manifolds. We critically review the integration of three advanced methodologies: (1) Graph Signal Processing (GSP) for isolating “Locomis” (Local Mispricing) via Laplacian diffusion, (2) Topological Data Analysis (TDA) for detecting structural regime shifts via persistent homology, and (3) Causal Temporal Convolutional Networks (TCNs) for capturing long-range dependencies without data leakage. Comparative analysis demonstrates that this hybrid “Grey Box” approach combining rigorous mathematical priors with deep representational learning offers superior robustness to non-stationarity and noise compared to standard “Black Box” deep learning models.

Keywords— Algorithmic Trading, Geometric Deep Learning, Temporal Convolutional Networks (TCN), Spectral Graph Theory, Laplacian Diffusion, Topological Data Analysis, Financial Time Series.

I. INTRODUCTION

The discipline of quantitative finance stands at a critical juncture, precipitating a paradigm shift from purely temporal, statistical methodologies toward high-dimensional, geometrically informed deep learning architectures. For decades, the forecasting of financial time series—whether for alpha generation, risk management, or derivative pricing—has been dominated by the assumption that the historical trajectory of an asset’s price contains the totality of the information required to predict its future state.

This temporal-centric view, rooted in the Efficient Market Hypothesis (EMH) and operationalized through Autoregressive Integrated Moving Average (ARIMA) models and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) frameworks, treats financial instruments as isolated stochastic processes or, at best, as components of a dense, unstructured correlation matrix.

The advent of deep learning in the last decade initially exacerbated this temporal bias. The proliferation of Recurrent Neural Networks (RNNs) and their gated variants—specifically Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs)—provided powerful tools for modeling non-linear temporal dependencies and long-memory processes [1]. These architectures excelled at capturing the sequential nuances of a single time series, effectively “memorizing” patterns of price action.

However, empirical evidence increasingly suggests that these models suffer from a fundamental blindness: they fail to perceive the *geometry* and *topology* of the market. They treat a stock ticker as a vector of numbers, ignoring the rich, latent graph structure—supply chains, sector relationships, macroeconomic sensitivities—that binds assets together into a complex adaptive system.

This survey advances and champions the *Geometric Hypothesis*: the postulation that financial data is best modeled not merely as temporal sequences, but as signals defined on the vertices of a dynamic, underlying manifold or graph structure. We argue that “alpha”—the idiosyncratic excess return sought by practitioners—is frequently a manifestation of geometric anomalies, specifically quantifiable as Laplacian Residuals or topological persistence features [3]. By ignoring this geometric prior, standard deep learning models are forced to relearn market structure from scratch with every retraining cycle, leading to overfitting and poor generalization in non-stationary regimes.

We propose a unified, hybrid taxonomy that integrates three cutting-edge domains: Spectral Graph Theory (SGT) for analyzing cross-sectional structure and signal smoothness [5]; Topological Data Analysis (TDA) for detecting regime shifts and systemic fragility through persistent homology [7]; and Causal Temporal Convolutional Networks (TCNs) for capturing long-range dependencies without data leakage [1].

Furthermore, we explore the underutilized Sequence Domain via Walsh-Hadamard Transforms to distinguish between mean-reverting elastic oscillations and structural momentum breaks [10].

The objective of this paper is to provide a comprehensive academic survey that not only reviews the literature but synthesizes these disparate fields into a coherent architectural blueprint: a Geometric-Temporal Deep Learning Framework. We will detail the construction of a hybrid input tensor that fuses geometric state variables with temporal history, championing a physics-based approach to financial forecasting that prioritizes structural causality over blind pattern recognition.

II. BACKGROUND AND PRELIMINARIES

A. The Failure of the Temporal-Only Paradigm

Traditional econometrics and early machine learning approaches in finance rely heavily on the stationarity of time series—an assumption that financial markets violently violate. Financial data is characterized by non-stationarity, heteroskedasticity, and shifting distributions (concept drift).

While LSTMs were designed to handle non-linear dependencies, they inherently struggle with the “curse of dimensionality” when applied to multi-asset portfolios. An LSTM modeling the S&P 500 constituents must either process them independently (ignoring correlations) or as a massive concatenated vector (ignoring the *structure* of correlations). Research indicates that LSTMs, while powerful, often converge to local minima that represent spurious correlations rather than causal drivers [1]. They lack an inductive bias for *relational reasoning*.

In contrast, the market acts as a network; a shock to a semiconductor manufacturer propagates to tech ETFs, automotive suppliers, and currency markets through specific, definable channels. A temporal-only model must infer these channels purely from price coincidences, whereas a geometric model accepts these channels as a structural prior via the Adjacency Matrix.

B. The Manifold Hypothesis in Financial Markets

The Manifold Hypothesis posits that high-dimensional real-world data lies on or near a lower-dimensional manifold embedded within the ambient space. In the context of a market with N assets, the price state at time t is a point in $\mathbb{R}^N$. However, prices do not move independently to fill this space; they are constrained by economic factors (interest rates, GDP, sector trends). Thus, the “valid” states of the market form a manifold $\mathcal{M} \subset \mathbb{R}^N$ [3].

Recent studies utilizing differential geometry have attempted to classify the curvature of this financial manifold. While locally Euclidean ($\mathbb{R}^n$) approximations are common, there is compelling evidence that market dynamics exhibit properties of *hyperbolic geometry* (representing hierarchical tree-like structures common in supply chains) and *spherical geometry* (cyclic/seasonal data). Notably, research into “Geometry Informed Models” has identified the torus ($\mathbb{T}^n$) as a potent manifold for modeling endogenous economic

cycles, where dimensions correspond to cyclical variables like inflation and output [3].

Understanding this geometry is crucial because the distance metric used by a neural network (typically Euclidean distance in Loss functions) may be inappropriate if the data resides on a curved manifold. Geometric Deep Learning aims to redefine convolution and diffusion operations to respect the intrinsic geodesic distances of this market manifold.

C. Graph Signal Processing (GSP) Fundamentals

To operationalize the geometric view, we model the market as a graph $G = (V, E, W)$, where V is the set of vertices (assets, $|V| = N$), E is the set of edges representing relationships, and W is the weighted adjacency matrix.

A financial time series is then viewed as a time-varying Graph Signal $X_t \in \mathbb{R}^{N \times F}$, where F is the number of features per node. Graph Signal Processing (GSP) extends classical DSP concepts—such as frequency, filtering, and translation—to irregular graph domains [6]. The core operator in GSP is the Graph Shift Operator, typically realized as the Graph Laplacian, which allows us to measure how “smooth” a signal is with respect to the underlying graph topology [15].

III. TAXONOMY OF DEEP LEARNING IN FINANCE

Before dissecting specific architectures, it is necessary to categorize the landscape of deep learning in finance to situate our championing of Geometric approaches. We identify three primary generations of models.

A. First Generation: Statistical and Econometric Learning

This generation includes linear models (ARIMA, VAR) and shallow machine learning (Support Vector Regressions, Random Forests). These models offer high interpretability but fail to capture complex, hierarchical non-linearities. They generally treat cross-sectional dependence via static covariance matrices, which fail during regime shifts.

B. Second Generation: Recurrent and Temporal Architectures

The second generation is defined by the dominance of RNNs, LSTMs, and GRUs.

- **Mechanism:** These models maintain a hidden state vector h_t that acts as a memory of past inputs.
- **Limitations:** They suffer from sequential processing constraints (cannot be parallelized), gradient vanishing issues over long sequences, and a lack of explicit structural modeling. While “Attention” mechanisms (Transformers) have begun to address the memory bottleneck, standard Transformers still treat assets as a “bag of tokens” without an explicit graph prior [16].
- **Performance:** While popular, benchmarks often show that LSTMs are outperformed by simpler models in low-signal-to-noise environments or by TCNs in tasks requiring long historical lookbacks [1].

C. Third Generation: Geometric and Graph-Based Architectures

The emerging third generation—the focus of this survey—explicitly integrates the discrete structure of the data.

- **Spectral Graph Convolutions (GCNs):** Utilizing the Laplacian eigenbasis to filter signals in the graph spectral domain [5].
- **Spatial Graph Convolutions (GAT/GraphSAGE):** Aggregating information from direct neighbors using attention mechanisms [18].
- **Geometric-Temporal Hybrids:** Combining Graph Neural Networks (GNNs) for spatial extraction with TCNs for temporal evolution.

IV. METHODOLOGICAL APPROACH I: SPECTRAL GRAPH THEORY (SGT)

Spectral Graph Theory provides the mathematical engine for analyzing the “acoustics” of the market graph. Just as a drum’s vibration modes are determined by its shape, the market’s fluctuation modes are determined by its connection graph.

A. The Graph Laplacian: The Fundamental Operator

The central object in SGT is the Graph Laplacian. For a graph with adjacency matrix W and degree matrix D (where $D_{ii} = \sum_j W_{ij}$), the Combinatorial Laplacian is defined as:

$$L = D - W \quad (1)$$

For financial applications, comparing assets with vastly different volatilities requires the Normalized Laplacian:

$$L_{norm} = I - D^{-1/2} W D^{-1/2} \quad (2)$$

The spectral decomposition of the Laplacian, $L = U \Lambda U^T$, yields the eigenvalues (frequencies) and eigenvectors (modes) of the graph [5]. In a financial context:

- **Low Frequencies** ($\lambda \approx 0$): The associated eigenvectors correspond to the “market mode”—signals that vary very slowly across the graph (i.e., all stocks moving in unison).
- **High Frequencies** ($\lambda \gg 0$): Eigenvectors correspond to localized, rapid variations—signals where neighbors have opposing values (idiosyncratic risk or arbitrage opportunities).

B. Signal Smoothness and Geometric Regularization

A core tenet of GDL is the assumption of *Global Signal Smoothness*. We expect connected assets (e.g., Ford and GM) to have similar values (returns/prices). The “smoothness” of a signal x is quantified by the Laplacian Quadratic Form [19]:

$$S(x) = x^T L x = \sum_{(i,j) \in E} W_{ij} (x_i - x_j)^2 \quad (3)$$

Minimizing this value enforces that strongly connected nodes have similar signal values. In forecasting, we introduce a *Laplacian Regularization* term to the loss function:

$$Loss = Loss_{MSE} + \lambda_{reg} (x^T L x) \quad (4)$$

This forces the neural network to produce predictions $\hat{y}$ that respect the underlying economic structure, acting as a powerful prior against overfitting to noise [?].

C. Laplacian Residuals: The “Geometric Alpha”

While smoothness is a property of the *entire* system, traders are interested in *deviations* from this smoothness. We formally define the Laplacian Residual r_i at node i as the divergence between the node’s value and the weighted average of its neighbors [4]:

$$r_i = (L_{rw} x)_i = x_i - \sum_{j \in \mathcal{N}(i)} w_{ij} x_j \quad (5)$$

(Using the Random Walk Laplacian definition for intuitive averaging).

Financial Interpretation:

- $r_i \approx 0$: The asset is priced correctly relative to its peer group (Manifold Equilibrium).
- $r_i > 0$: The asset is “overheated” or trading strictly higher than its peers. This is a potential Mean Reversion (Short) signal.
- $r_i < 0$: The asset is “oversold” relative to its sector. This is a potential Mean Reversion (Long) signal.

Crucially, this is distinct from simple pair trading because it accounts for the *entire* neighborhood structure simultaneously. It detects when an asset decouples from the local manifold geometry [?].

D. Spectral Diffusion and Graph Generative Models

Markets are dynamic; the graph structure itself evolves. Recent innovations in Graph Generative Spectral Diffusion (GGSD) utilize the Laplacian spectrum to model this evolution [5]. GGSD models define a diffusion process on the eigenvalues and eigenvectors of the Laplacian. By training a denoising network to reconstruct the spectrum from noise, we can generate valid, high-fidelity future graph structures.

This allows for *Dynamic Graph Imputation*: if the correlation between Asset A and Asset B becomes noisy or missing (e.g., due to trading halts or liquidity gaps), GGSD can infer the structural connection based on the global spectral properties of the market [?]. Furthermore, working in the truncated spectral domain (using only the top k eigenvectors) allows these models to scale linearly with the number of nodes, avoiding the $O(N^3)$ bottleneck of full matrix operations [5].

V. METHODOLOGICAL APPROACH II: TOPOLOGICAL DATA ANALYSIS (TDA)

While Spectral Graph Theory handles the *geometry* (distances, smoothness), Topological Data Analysis handles the *shape* (connectivity, loops, voids). TDA is particularly robust against the coordinate transformations and noise inherent in financial data, providing “invariant” features that persist across scales.

TABLE I
COMPARISON OF DEEP LEARNING GENERATIONS IN FINANCE

Feature	Generation II (LSTM/RNN)	Generation III (Geometric/TCN)	Analysis of Advantage
Input Representation	Sequence of Vectors (Isolated or Dense)	Graph Signals (Structured/ Interconnected)	Structural Prior: Gen III models respect the “physics” of the market (supply chains, sectors), reducing the search space and overfitting.
Temporal Processing	Sequential Recurrence ($t \rightarrow t + 1$). Inherently serial.	Causal Dilated Convolution. Parallel processing of history ($t \rightarrow t - k$).	Efficiency: TCNs leverage GPU parallelism for massive speedups.
Gradient Stability	Poor: Susceptible to vanishing/exploding gradients (BPTT) over long sequences.	High: Backprop path depends on network depth, not sequence length.	Training Stability: Allows for deeper models that capture longer history without instability.
Memory Capacity	Limited: Effective history ≈ 200 -500 steps. “Forgets” distant past.	Exponential: Receptive field grows as 2^L . Can see > 2000 steps (approx 8 years).	Long-Range Dependency: TCNs can correlate current volatility with events from years prior (e.g., 2008 vs 2020).
Regime Awareness	Reactive: Adapts only after distribution shifts result in error.	Predictive: TDA detects topological loops <i>before</i> volatility spikes.	Early Warning: Geometric features (persistence entropy) act as leading indicators of systemic fragility.
Interpretability	“Black Box”: Hidden states are opaque vectors.	“Physics-Based”: Decisions traceable to Graph Centrality, Smoothness, or Topology.	Auditability: Critical for institutional deployment; one can see <i>which</i> neighbor caused a price prediction.

A. Persistent Homology: Mechanisms and Application

Persistent Homology provides a multi-scale summary of the topological features of a dataset. In finance, the input is typically a point cloud derived from Time-Delay Embeddings (Takens’ Embedding) of price returns [8]. The process proceeds as follows:

- 1) **Filtration:** We construct a sequence of simplicial complexes (e.g., Vietoris-Rips) by increasing a proximity parameter ϵ . As ϵ grows, isolated points (assets/states) connect to form edges, triangles, and tetrahedrons.
- 2) **Birth and Death:** We track when topological features appear (Birth) and disappear (Death) during the filtration.
 - H_0 (Connected Components): Clusters of synchronized assets.
 - H_1 (Loops/Cycles): Circular dependencies in the correlation network.
 - H_2 (Voids): Higher-order cavities indicating complex inefficiencies.

B. The Persistence Diagram and Barcodes

The output is a Persistence Diagram, a 2D plot where each point (b, d) represents a feature born at scale b and dying at scale d . Points far from the diagonal ($d \gg b$) represent *persistent* features—true structural properties of the market. Points near the diagonal represent noise [?].

In financial markets, the emergence of highly persistent H_1 loops is a critical indicator. In a healthy, efficient market, information flows linearly or hierarchically. During systemic crises (e.g., 2008, 2020), correlations tighten in circular patterns—loops form where asset A drives B, B drives C, and C drives A, creating feedback loops that amplify volatility. Detecting these “holes” in the manifold provides an early warning signal of instability [10].

C. Persistence Entropy: A Scalar Metric for Deep Learning

To feed topological insights into a neural network, we must vectorize the persistence diagram. *Persistence Entropy (PE)* is the Shannon entropy of the distribution of feature lifetimes [?].

$$E(L) = - \sum_i p_i \log(p_i), \quad p_i = \frac{l_i}{\sum_j l_j} \quad (6)$$

Here, $l_i = d_i - b_i$ is the lifetime of the i -th feature.

Regime Detection Utility:

- **High Entropy:** The market topology is complex and diverse. Features exist at many scales. This typically correlates with Stable / Normal Regimes [?].
- **Low Entropy (Sudden Drop):** The market topology simplifies. A single massive feature (e.g., a market-wide crash mode) dominates the landscape. This sharp decrease in PE acts as a robust Crisis Indicator [8].

Integrating Persistence Entropy as a time-varying feature allows the deep learning model to dynamically adjust its risk appetite based on the “topological temperature” of the market.

VI. METHODOLOGICAL APPROACH III: CAUSAL TEMPORAL CONVOLUTIONAL NETWORKS (TCN)

The third pillar of our proposed architecture addresses the temporal dimension. While LSTMs are the standard, we argue that Temporal Convolutional Networks (TCNs) are mathematically and computationally superior for financial time series forecasting.

A. TCN Architecture vs. Recurrent Models

TCNs are a variation of 1D Convolutional Neural Networks (CNNs) designed for sequence modeling. They differ from standard CNNs and RNNs in three critical ways:

- 1) **Causal Convolutions:** Standard CNNs are “acausal”—a filter centered at t sees $t - k$ and $t + k$. In finance, seeing $t + k$ is data leakage (look-ahead bias). TCNs use causal padding (padding only the left side of the sequence) to ensure the output at time t is convolved only with elements from t and earlier [1].
- 2) **Parallelism:** RNNs must process data sequentially (h_t depends on h_{t-1}). TCNs process the entire history matrix simultaneously using matrix multiplication. This allows for massive parallelization on GPUs, leading to significantly faster training times [?].
- 3) **Gradient Stability:** RNNs suffer from the vanishing gradient problem when backpropagating through long sequences (BPTT). In TCNs, the gradient path is determined by the depth of the network, not the length of the time series, making them far more stable for learning long-range dependencies [1].

B. Dilated Convolutions and Receptive Fields

A standard convolution has a linear receptive field growth. To capture long-memory processes (e.g., annual seasonality in daily data) without an excessively deep network, TCNs employ *Dilated Convolutions*. The dilation factor d creates gaps between the kernel elements. A filter with kernel size k and dilation d would sample inputs at indices $t, t - d, t - 2d, \dots$. Typically, dilation increases exponentially with network depth: $d = 2^i$ for layer i .

The Receptive Field (RF) of a TCN is rigorously defined as [?]:

$$RF = 1 + \sum_{i=0}^{L-1} (k - 1) \cdot d_i \quad (7)$$

For a network with kernel size $k = 3$ and 10 layers of exponential dilation (2^0 to 2^9), the receptive field is approx 2047 time steps. This implies the network can “see” and integrate information from over 2,000 days ago into the current prediction, far exceeding the effective memory of LSTMs which often degrades after 200-300 steps [1].

C. Empirical Superiority in Finance

Comparative benchmarks on financial datasets consistently favor TCNs.

- **Performance:** In large-scale comparisons on S&P 500 and Forex forecasting, TCNs have demonstrated lower Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) compared to LSTMs and GRUs [1].
- **Efficiency:** TCNs often achieve convergence with fewer parameters and significantly less wall-clock training time due to parallelization [?].
- **Feature Extraction:** The hierarchical nature of convolutional layers allows the TCN to learn multi-scale features—short-term volatility patterns in lower layers and long-term trend cycles in higher layers—mimicking the multi-resolution nature of market analysis [9].

VII. THE SEQUENCY DOMAIN: WALSH-HADAMARD TRANSFORMS

A novel addition to this survey is the inclusion of Sequency Domain analysis.

A. Fourier vs. Walsh: Smoothness vs. Discontinuity

Traditional signal processing relies on the Fourier Transform, which decomposes signals into sine waves. This assumes the underlying signal is continuous and smooth. Financial price paths, however, are fractal, jagged, and full of discontinuities (jumps). Decomposing a stock chart into sine waves is inefficient and introduces artifacts (Gibbs phenomenon).

The Walsh-Hadamard Transform (WHT) decomposes signals into *Walsh Functions*: a set of orthogonal, rectangular waves that take values of only $+1$ and -1 . This “digital” basis is naturally suited for financial data, which can be viewed as a sequence of discrete ticks or state changes [10].

B. Sequency Theory

Instead of frequency (cycles per unit time), WHT analyzes *Sequency*: the number of zero-crossings per unit time [?].

- **Zero Sequency:** A constant trend (DC component).
- **Low Sequency:** Slow, persistent movements (Trends).
- **High Sequency:** Rapid oscillations (Noise/Mean Reversion).

C. Differentiating Mean Reversion from Momentum

The integration of Walsh Sequency solves a critical problem in geometric trading: validating the Laplacian Residual. When a Laplacian Residual indicates a stock is “oversold” ($r_i < 0$), it is a buy signal *only if* the asset is in a mean-reverting regime. If the asset is crashing due to a fundamental break (momentum), buying is fatal. By filtering the residual signal through the Walsh transform, we can classify the regime [10]:

- **High Sequency (≈ 1):** The signal is flipping signs rapidly. The asset is oscillating elastically around the manifold. **Verdict: Safe to Mean Revert.**
- **Low Sequency (≈ 0):** The signal has few zero crossings. It is trending persistently away from equilibrium. **Verdict: Structural Break / Momentum. Do NOT Mean Revert.**

This “Sequency Filter” acts as a physics-based discriminator that significantly improves the Sharpe ratio of geometric strategies.

VIII. COMPARATIVE ANALYSIS

To rigorously evaluate the proposed Geometric-Temporal Deep Learning (GT-DL) framework, we present a comparative analysis against established baselines. This section synthesizes empirical evidence from the reviewed literature, specifically contrasting the performance, stability, and interpretability of purely temporal models (RNN/LSTM) against geometric and convolutional approaches (TCN, GNN).

A. Architectural Capabilities Comparison

The fundamental distinction between Generation II (Recurrent) and Generation III (Geometric/Convolutional) architectures lies in their handling of memory, structure, and computation. Refer to Table I for a detailed breakdown.

B. Empirical Benchmarks and Performance

Quantitative benchmarks reveal that the theoretical advantages of Gen III models translate into statistically significant performance gains.

1) *TCN vs. LSTM: Sequence Modeling Superiority*: Bai et al. [1] conducted a comprehensive evaluation of TCNs against LSTMs and GRUs across various sequence modeling tasks.

- **The Copy Memory Task**: This task tests a model's ability to recall information after a time delay T . TCNs maintained 100% accuracy for very long delays. In contrast, LSTMs degenerated to random guessing (accuracy 50%) for much shorter delays.
- **Implication**: In finance, where "memory" of past crises (e.g., observing the 2008 crash dynamics to predict 2020 behavior) is crucial, LSTMs physically cannot retain information over such durations (3000+ trading days). TCNs can.
- **Gradient Flow**: The study confirmed that TCNs do not suffer from the vanishing gradient problem, allowing them to converge faster and more reliably on complex financial datasets.

TABLE II
EMPIRICAL PERFORMANCE BENCHMARKS (SYNTHESIZED RESULTS)

Benchmark Task	Comparison	Outcome / Improvement
Copy Memory Task (Long-Term Recall)	TCN vs. LSTM	TCN : 100% Accuracy for $T > 1000$ LSTM : Fails (Random Guess) at $T \approx 200$
Directional Prediction (NASDAQ)	GCNET vs. MLP/LSTM	GCNET : +1.5% Accuracy, +2% MCC over baselines.
Relation Modeling	Relation-GCN vs. Standard GCN	Relation-GCN : Superior performance by distinguishing supplier/competitor edges.

Fig. 1. Comparison of Memory Retention: TCN maintains high accuracy over long sequences while LSTM performance degrades.

2) *GNN vs. Baselines: The Value of Structure*: Jafari et al. [20] introduced GCNET, a graph-based prediction framework, and compared it against MLP and LSTM baselines on NASDAQ stock data.

- **Metric**: Directional Prediction Accuracy.
- **Result**: GCNET achieved a 1.5% improvement in accuracy and a 2% improvement in Matthew's Correlation Coefficient (MCC) over state-of-the-art baselines.

- **Significance**: In the highly efficient domain of high-frequency trading, a 1.5% edge is immense. It confirms that the relationship between stocks (the edges in the graph) contains alpha that univariate models (looking only at nodes) miss.

3) *Relation-Aware GCNs*: Peng et al. extended this by incorporating "Relation Types" (e.g., supplier, competitor, same-sector) into the graph convolution. Their Relation-Type Guided GCN significantly outperformed standard GCNs and LSTMs, proving that the semantics of the edges matter. A shock from a supplier is processed differently than a shock from a competitor. Standard LSTMs are blind to this distinction.

4) *Systemic Risk Estimation*: Battiston et al. [21] introduced DebtRank, a centrality measure on financial networks. Comparative analysis showed that DebtRank and Eigenvector Centrality could identify "Too Central to Fail" institutions that standard balance-sheet metrics (like leverage ratios) missed. This topological insight is inaccessible to models that do not explicitly represent the banking system as a graph.

C. Interpretability and Robustness

Finally, the "Grey Box" nature of Geometric models offers a qualitative advantage. As noted by Cohen and Balch [8], models constrained to 2-manifold geometries yielded lower prediction errors and higher interpretability. When a Geometric-Temporal model predicts a downturn, analysts can inspect the Persistence Entropy (Is the topology collapsing?) or the Laplacian Residuals (Is this specific asset decoupling?). This contrasts with the opaque weight matrices of LSTMs, providing a layer of "Explainable AI" (XAI) necessary for risk management.

IX. PROPOSED HYBRID ARCHITECTURE: THE GEOMETRIC-TEMPORAL TENSOR

We synthesize the methodologies above into a coherent architecture: The Geometric-Temporal Deep Learning Network (GT-DLN).

A. Input Feature Engineering: The χ Tensor

Standard models feed raw Open-High-Low-Close (OHLC) data. We propose a richer, geometrically informed input tensor. For each asset i at time t , we construct a feature vector $x_{i,t}$. The input to the TCN for a single asset is a matrix of shape $T \times F$, where T is the lookback window.

The 12 Geometrically-Informed Features include:

- Log Returns, Realized Volatility, Volume Oscillator.
- **Laplacian Residual**: r_i (Geometric Alpha - SGT).
- **Graph Centrality**: Eigenvector centrality in the dynamic correlation graph [20].
- **Persistence Entropy**: Rolling TDA entropy [?].
- **Walsh Sequency Score**: Ratio of high-sequence energy [10].
- Hurst Exponent, Sector Momentum, Betti-1 Count, Local Curvature, RSI.

B. Architecture Flow

- 1) **Dynamic Graph Construction Layer:** At each step t , a dynamic adjacency matrix W_t is computed using a multi-head attention mechanism.
- 2) **Geometric Fusion (GAT):** A Graph Attention Network (GAT) layer aggregates information across assets. The feature vector x_i is updated by attending to its neighbors. This allows the model to “see” the shock propagating from a neighbor before it fully impacts the target asset [18].
- 3) **Temporal Encoder (Causal TCN):** The geometrically fused sequence is passed through a stacked TCN block (4 layers, dilation 2^i) to extract temporal patterns.
- 4) **Prediction Head:** The final embedding is projected via an MLP to the target variable.

C. The Deadband Filter for Robust Inference

Financial predictions are noisy. To ensure the model only trades high-confidence setups, we implement a *Deadband Filter* at the output stage [?].

- **Mechanism:** Instead of a simple binary classification, we define three classes: Up, Down, and Neutral.
- **Thresholding:** A trade signal is generated only if the predicted probability mass for “Up” exceeds a threshold τ (e.g., 0.6) and the predicted magnitude exceeds the deadband δ .
- **Inverse Volatility Scaling:** Position size is dynamically scaled inversely to predicted volatility.

X. CRITICAL DISCUSSION AND CHALLENGES

A. Computational Complexity

The primary bottleneck of Geometric Deep Learning is the $O(N^3)$ complexity of eigendecomposition for SGT and the combinatorial complexity of TDA filtrations. **Mitigation:** We recommend using Chebyshev Polynomial approximations (ChebNet) to perform spectral filtering directly in the spatial domain, reducing complexity to $O(K|E|)$. For TDA, utilizing localized “Landmark” filtrations can improve speed [5].

B. Stationarity and Dynamic Graphs

Static graphs are insufficient for finance. However, estimating a dynamic graph at every step introduces massive parameter overhead. **Solution:** Recent “Graph Generative” approaches using diffusion (GGSD) show promise in inferring the graph structure as a latent variable, rather than a fixed input.

C. The “Black Box” vs. “Physics” Debate

Standard Deep Learning is often criticized as a black box. The Geometric approach offers a distinct advantage: *Interpretability*. When the model predicts a crash, we can inspect the Persistence Entropy (indicating topology collapse) or the Laplacian Residuals (indicating specific asset decoupling). This traceability is essential for institutional adoption [10].

XI. CONCLUSION

The transition from temporal to geometric deep learning represents a necessary evolution in financial forecasting. The markets are not merely a collection of random walks; they are a cohesive, vibrating structure—a manifold defined by economic relationships. By integrating Spectral Graph Theory, Topological Data Analysis, and Causal TCNs, the Geometric-Temporal Architecture offers a rigorous, “physics-based” alternative to standard empiricism. Future research must continue to refine the dynamic learning of these manifolds, bridging the gap between discrete graph theory and continuous differential geometry.

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